SEMINAR AND PROJECT INITIATION COURSE

Course Code: 315003 | Semester V | MSBTE K Scheme

INDIVIDUAL SEMINAR TOPICS, PPT FORMAT AND REPORT FORMAT

Department: Diploma in Computer Engineering

1. Instructions to Students

- Each student must prepare and present one seminar individually. The topic must be finalised with the faculty guide.
- Select a topic from the allotment list below. The guide may approve a suitable modification or a closely related topic.
- Use authentic and updated information from textbooks, research papers, official documentation and reliable technical sources.
- The presentation must not exceed 20 minutes, including the question-and-answer session.
- The PPT must contain no more than 15 slides.
- Submit one printed hard copy of the seminar report and one soft copy containing the PPT and report.
- The report should normally remain within 25–30 pages and include only content relevant to the selected technology.
- Do not copy material blindly. Explain concepts in your own words and acknowledge every source of information, image, diagram and data.
- Seminar is an individual activity. Project initiation is a separate group activity; MSBTE suggests project groups of 3–4 students.

2. Individual Seminar Topic Allotment – 43 Students

Roll No	Student Name	Suggested Seminar Topic
1	MUJAVAR AMANUDDIN MAZHARUDDIN	Artificial Intelligence in Everyday Life
2	MUJAWAR MOIZODDIN MAZHARUDDIN	Generative AI and Large Language Models
3	ABDULLAH SAQUIB ABDUL HAFEEZ	Machine Learning: Concepts and Real-World Applications
4	ABDUR RAHMAN MUZAMMIL HUSAIN	Deep Learning and Artificial Neural Networks
5	ANSARI MOHD ADNAN KHURSHEED AALAM	Computer Vision and Image Recognition
6	AREEB AHMED LAEEQUE AHMED	Face Recognition Technology and Its Applications
7	AZQA FATIMA KHALID	AI-Based Attendance Monitoring System
8	CHOWDHARY RUKAYYA SHAMIM AHMAD	Internet of Things (IoT) and Smart Devices
9	HANZALA ZAHID AHMAD	Smart Home Automation Using IoT
10	HIRAY SHIVAM SHASHIKANT	Cloud Computing: IaaS, PaaS and SaaS
11	LOOMWALA FATEMA HAIDAR ALI	Microsoft Azure Cloud Services
12	MANTASHA RAFEEQUE AHMED	Amazon Web Services (AWS) Fundamentals
13	MASEERA IRAM MOHAMMAD ISMAIL	Edge Computing and Its Applications

14	MOHAMMAD OWAIS MOHAMMAD SHAREEF	Cybersecurity Threats and Prevention
15	MOHAMMAD REHAN MOHAMMAD ISMAIL	Ethical Hacking and Penetration Testing
16	MOHAMMED UMER MOHAMMED SHABAN	Blockchain Technology and Applications
17	MOHAMMAD UMAIR MOHAMMAD AYAZ KHAN	Cryptocurrency and Digital Wallet Security
18	MOHAMMED USAID MUBEEN AHMED	Database Management Systems: SQL vs NoSQL
19	MOMIN AAFREEN FIRDAUS ZAKIR HUSAIN	Big Data and Data Analytics
20	MOMIN AFFAN MALIK ANEES AHMED	Data Science Process and Applications
21	MOMIN MARIYAM ZUBAIR AHMED	Python Programming for Automation
22	MOMIN NABEELA ZIYAUDDIN	Web Development: Front End and Back End
23	RATLAMWALA BURHANUDDIN ABBAS	Progressive Web Applications (PWA)
24	SHAH NABEEL AHMED ABDUL SATTAR	Application Programming Interfaces (APIs) and REST
25	SHAIKH KAFEEL AHMED AQUEEL AHMED	Mobile Application Development Technologies
26	SHAIKH SAARA NAAZ KADIR	DevOps and CI/CD Pipeline
27	SHAIKH TANZEELA NAAZ ARIF	Docker and Containerisation
28	TUFAIL ANJUM RIZWAN AHMAD	Virtualisation and Virtual Machines
29	YUSUF REJA	Computer Networks: IPv4, IPv6 and Subnetting
30	KRISHNAKANT PAWAN JADHAV	Wi-Fi 6 and Modern Wireless Networking
31	MD MAHIN	5G Technology and Its Applications
32	NARJIS BANO AFZAL BEG	QR Code Technology and Secure Digital Services
34	MD SAHANBAJ	Biometric Authentication Systems
35	MOHAMMAD SAAD	Augmented Reality (AR) and Virtual Reality (VR)
36	ISRA JAMEELURRAHMAN	Robotics and Robotic Process Automation (RPA)
37	MOHAMMED AZHAR AKHTAR	Natural Language Processing (NLP)
38	KHAN NASHRAH KARIM	Recommendation Systems Used by Online Platforms
39	USHIRE GANESH TATYA	Software Testing and Automation Testing
40	YUSUFI ABDULLAH YUSUFI MUJEEB.	Git and GitHub for Collaborative Software Development
41	SHAIKH DANIYAL ZAHEER ABBAS	Digital Payments and UPI Technology
42	ZURAIZ AADIL	E-Governance and Digital India Services
43	SHAH SAMEER	Green Computing and Sustainable IT
44	ANSARI MD SHAFEY	Low-Code and No-Code Application Development

3. Seminar PPT Format (Maximum 15 Slides)

Slide	Slide Title	Required Content
1	Title Slide	Seminar title; student name; roll/enrolment number; Diploma in Computer Engineering; Semester V; institute name and logo; academic year; guide name.

2	Introduction	Brief introduction, background and why the topic is important.
3	Problem / Need / Motivation	What problem does the technology address? Why is it required?
4	Objectives and Scope	3–5 clear objectives and boundaries of the seminar.
5	Basic Concepts / Terminology	Important terms, definitions and fundamentals.
6	Technology / Architecture	Main components, architecture or block diagram.
7	Working Principle / Process	Step-by-step working, workflow or lifecycle.
8	Tools, Platforms and Technologies	Relevant software, hardware, platforms, languages or standards.
9	Applications / Real-World Use Cases	Practical applications, preferably with industry or local examples.
10	Case Study / Example	One real implementation, product, organisation or illustrative example.
11	Advantages	Technical, operational, economic or user benefits.
12	Limitations / Challenges	Disadvantages, risks, privacy/security issues, cost or implementation challenges.
13	Future Scope / Emerging Developments	Recent developments and likely future applications.
14	Conclusion and References	Key learning points and 3–5 important references.
15	Thank You / Q&A	Thank the audience and invite questions.

4. PPT Design and Typographical Guidelines

- Use English (British) and maintain consistent formatting throughout the presentation.
- MSBTE syllabus typography: Times New Roman; body text 12 pt; subsection headings 12 pt bold; section headings 12 pt bold capitals; chapter/topic title 14 pt capitals.
- Keep slides readable: use short bullet points, meaningful diagrams, labelled screenshots and tables. Avoid paragraphs copied from the report.
- Use consistent colours and a simple background. Ensure adequate contrast and readable diagrams.
- Put the candidate's name, diploma/department, year of submission and institute name on the first slide.
- Suggested delivery: approximately 12–15 minutes for presentation and 3–5 minutes for questions, within the 20-minute limit.

5. Seminar Report Format (25–30 Pages Maximum)

Arrange the report in the following order. The first three formal pages should follow the MSBTE annexure formats where provided by the institute.

Section	**What to include**
Cover Page	Use the MSBTE Annexure I layout: MSBTE and institute logos; SEMINAR REPORT; seminar title; programme; student name and enrolment number; academic year; institute name.
Title Page	Repeat the seminar title and student/institute details in the institute-prescribed title-page layout.
Certificate by Guide	Certificate signed by the internal guide and other institute authorities as prescribed by the institute.
Acknowledgement	Briefly acknowledge the guide, department, institute and other people who supported the seminar.
Abstract (one page)	Summarise the purpose, topic, approach/methodology (if any), key concepts and conclusion.

Index / Table of Contents	List chapters, sections and page numbers.
List of Figures	List figure numbers, captions and page numbers. Add a list of tables if required.
Chapter 1 – Introduction	Background, overview, need/motivation, problem or context, objectives and scope of the seminar.
Chapter 2 – Literature Review	Summarise relevant books, research papers, articles, standards and official technical documentation. Mention what each source contributes.
Chapter 3 – Technology / Concepts	Explain the selected technology, terminology, components, architecture, tools/platforms and working principle.
Chapter 4 – Applications / Case Study	Discuss practical applications, industry use cases and one suitable case study or example.
Chapter 5 – Advantages and Disadvantages	Present benefits and limitations objectively. Include a comparison table where useful.
Chapter 6 – Future Scope and Conclusion	Discuss future developments and conclude with the main learning outcomes.
Project Initiation (brief section)	Include the problem statement, scope, objectives, stakeholders, and identification of platform/equipment/resources. This is the initial project-planning component, not a claim that a complete project has been implemented.
Bibliography / References	List all books, papers, websites, official documentation and other sources actually used. Include author/organisation, title, year and URL/DOI/access date where applicable.

6. Report Formatting Guidelines

- Use Times New Roman, 12 pt for body text; subsection headings 12 pt bold; section headings 12 pt bold capitals; chapter/topic titles 14 pt capitals, as specified in the syllabus.
- Use justified alignment for body paragraphs. Keep heading styles, numbering and spacing consistent.
- Number chapters and sections systematically (for example, 1, 1.1, 1.2). Give every figure and table a number and caption.
- Use clear, readable diagrams. Mention the source below any borrowed figure, table or image.
- Keep the report within 25–30 pages. Include only relevant technology, platform, operating system and tools.
- Use a consistent reference style throughout. Do not list sources that were not consulted.

7. Soft Copy Folder Structure

- Seminar_Submission_[RollNo]_[StudentName]/
- Presentation/ → Seminar PPT (maximum 15 slides)
- Documentation/ → Seminar report (Word and/or PDF as instructed)
- Check the soft copy for viruses before submission. MSBTE specifies that the soft copy should be included on the back cover of the report in a clear packet.

8. Seminar Assessment – MSBTE Rubrics

Formative Assessment (PA): the syllabus gives the following suggested criteria, totalling 50 marks and converted proportionately to 25 marks.

Sr.	Criteria	Marks

No.		
1	Selection of topic/theme of seminar	5
2	Literature review and data presentation	5
3	Quality of preparation and innovativeness	5
4	Q&A handling	5
5	Time management	5
6	Seminar presentation report	10
		Total: 35*